SEMTM0051 — Foundations of Artificial Intelligence

Backpropagation

Instructor: Roberto Colomboni version of 27 October 2026

6 Exercises

The exercise numbering is the same as in the lecture notes dated 21 October 2026. References to earlier equations refer to those notes.

The following exercises practise a complete forward and backward pass for regression, binary classification and classification with three classes. Use the formulas from the main text. The optional appendix is not required. Keep every parameter at its forward-pass value until all the gradients have been calculated.

1. A complete numerical backward pass. Use the Lecture 4 network

$$\begin{aligned} \boldsymbol{a}_1(\boldsymbol{x}) &= \boldsymbol{x} \cdot \boldsymbol{W}_1 + \boldsymbol{b}_1, \qquad & \boldsymbol{\phi}_1(\boldsymbol{x}) &= \mathbf{ReLU}(\boldsymbol{a}_1(\boldsymbol{x})), \\ f_{\boldsymbol{\theta}}(\boldsymbol{x}) &= \boldsymbol{\phi}_1(\boldsymbol{x}) \cdot \boldsymbol{W}_2 + b_2. \end{aligned}$$

with

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 2 & -1 \\ -1 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{b}_1 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}, \qquad \boldsymbol{W}_2 = \begin{pmatrix} 2 \\ -1 \\ 1/2 \end{pmatrix}, \quad b_2 = -1.$$

Take $\boldsymbol{x} = (2, 0)$ and the regression target $y = 1$. For this fixed observation, the objective is $\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = (f_{\boldsymbol{\theta}}(\boldsymbol{x}) - y)^2$. The final preactivation row is $\boldsymbol{a}_2(\boldsymbol{x}) = (f_{\boldsymbol{\theta}}(\boldsymbol{x}))$.

(a) Compute $\boldsymbol{a}_1(\boldsymbol{x})$, $\boldsymbol{\phi}_1(\boldsymbol{x})$, $f_{\boldsymbol{\theta}}(\boldsymbol{x})$, and the squared loss. State the shapes of both weight matrices and the hidden rows.

(b) Write the remaining-loss functions $J_1 : \mathbb{R}^{1\times 3} \to \mathbb{R}$ and $J_2 : \mathbb{R}^{1\times 1} \to \mathbb{R}$ for this network. Calculate $\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x}))$ and then $\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x}))$. Use these rows to find the gradients of $\mathcal{L}_{\text{sq}}$ with respect to every weight and bias, evaluated at $\boldsymbol{\theta}$. Keep every parameter at its forward-pass value throughout this calculation.

(c) Explain how the negative third preactivation affects the backward pass. Explain separately why the second row of the $\boldsymbol{W}_1$ gradient is zero for this observation.

(d) Only after completing the backward pass, write one gradient descent update of all parameters using a learning rate $\eta > 0$.

2. Binary classification. Consider a network with one input coordinate, two ReLU hidden units and one output logit:

$$\begin{aligned} \boldsymbol{a}_1(\boldsymbol{x}) &= \boldsymbol{x} \cdot \boldsymbol{W}_1 + \boldsymbol{b}_1, \qquad & \boldsymbol{\phi}_1(\boldsymbol{x}) &= \mathbf{ReLU}(\boldsymbol{a}_1(\boldsymbol{x})), \\ f_{\boldsymbol{\theta}}(\boldsymbol{x}) &= \boldsymbol{\phi}_1(\boldsymbol{x}) \cdot \boldsymbol{W}_2 + b_2. \end{aligned}$$

Use

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 2 \end{pmatrix}, \quad \boldsymbol{b}_1 = \begin{pmatrix} 0 & 0 \end{pmatrix}, \qquad \boldsymbol{W}_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad b_2 = 0,$$

with input row $\boldsymbol{x} = (1) \in \mathbb{R}^{1\times 1}$ and target $y = 1$. The predicted class-1 probability and the loss for this fixed observation are

$$p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(f_{\boldsymbol{\theta}}(\boldsymbol{x})), \qquad \mathcal{L}_{\log}(\boldsymbol{\theta}) = \ell_{\log}(p_{\boldsymbol{\theta}}(\boldsymbol{x}), y).$$

As in Exercise 1, write the final preactivation as the row $\boldsymbol{a}_2(\boldsymbol{x}) = (f_{\boldsymbol{\theta}}(\boldsymbol{x}))$.

(a) Compute the hidden preactivations and activations, the output logit, the class-1 probability and the logistic loss. Keep the loss in exact form.

(b) Write $J_2 : \mathbb{R}^{1\times 1} \to \mathbb{R}$, including the sigmoid in this remaining function. Use (24) to calculate $\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x}))$, then find $\nabla_{\boldsymbol{W}_2}\mathcal{L}_{\log}(\boldsymbol{\theta})$ and $\frac{\partial \mathcal{L}_{\log}}{\partial b_2}(\boldsymbol{\theta})$.

(c) Calculate $\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x}))$ using the saved preactivations and $\boldsymbol{W}_2$. Use this row to find $\nabla_{\boldsymbol{W}_1}\mathcal{L}_{\log}(\boldsymbol{\theta})$ and $\nabla_{\boldsymbol{b}_1}\mathcal{L}_{\log}(\boldsymbol{\theta})$.

3. Classification with three classes. Keep the input and the first-layer parameters from Exercise 2. Replace the output layer by three logits, using

$$\boldsymbol{W}_2 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \boldsymbol{b}_2 = \begin{pmatrix} -1 & -1 & 2 \end{pmatrix}.$$

The network output and probability row are now

$$\begin{aligned} \boldsymbol{a}_2(\boldsymbol{x}) = \boldsymbol{f}_{\boldsymbol{\theta}}(\boldsymbol{x}) &= \boldsymbol{\phi}_1(\boldsymbol{x}) \cdot \boldsymbol{W}_2 + \boldsymbol{b}_2, \\ \boldsymbol{p}_{\boldsymbol{\theta}}(\boldsymbol{x}) &= \operatorname{softmax}(\boldsymbol{a}_2(\boldsymbol{x})). \end{aligned}$$

The class labels are $1, 2, 3$, and the observed class is $y = 2$. For this fixed observation, use the cross-entropy objective

$$\mathcal{L}_{\text{multi}}(\boldsymbol{\theta}) = \ell_{\text{multi}}(\boldsymbol{p}_{\boldsymbol{\theta}}(\boldsymbol{x}), y) = -\log p_{\boldsymbol{\theta},2}(\boldsymbol{x}).$$

You may reuse the hidden preactivations and activations from Exercise 2.

(a) Compute the three logits, their softmax probabilities and the cross-entropy loss. Keep the loss in exact form.

(b) Write $J_2 : \mathbb{R}^{1\times 3} \to \mathbb{R}$, including softmax in this remaining function. Use (25) to calculate $\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x}))$, then find $\nabla_{\boldsymbol{W}_2}\mathcal{L}_{\text{multi}}(\boldsymbol{\theta})$ and $\nabla_{\boldsymbol{b}_2}\mathcal{L}_{\text{multi}}(\boldsymbol{\theta})$.

(c) Compute $\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) \cdot \boldsymbol{W}_2^{\mathsf{T}}$. Write out the three contributions to its first entry before adding them. Apply the componentwise ReLU factors to obtain $\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x}))$, then calculate $\nabla_{\boldsymbol{W}_1}\mathcal{L}_{\text{multi}}(\boldsymbol{\theta})$ and $\nabla_{\boldsymbol{b}_1}\mathcal{L}_{\text{multi}}(\boldsymbol{\theta})$.

Solutions

As in the lecture notes, each J_r gives the loss obtained by starting from a possible preactivation at layer r and carrying out the remaining calculations. Its target and the parameters of later layers are held fixed while we differentiate its argument. All gradients below are evaluated at the parameter values used in the corresponding forward pass.

1. A complete numerical backward pass.

(a) Forward pass and dimensions. The forward pass gives

$$\begin{aligned} \boldsymbol{a}_1(\boldsymbol{x}) &= (2, 5, -2), \qquad \boldsymbol{\phi}_1(\boldsymbol{x}) = (2, 5, 0), \\ f_{\boldsymbol{\theta}}(\boldsymbol{x}) &= 2 \cdot 2 + 5 \cdot (-1) + 0 \cdot \tfrac{1}{2} - 1 = -2. \end{aligned}$$

Thus $\mathcal{L}_{\mathrm{sq}}(\boldsymbol{\theta}) = (-2-1)^2 = 9$. The shapes are 2×3 for $\boldsymbol{W}_1$, 3×1 for $\boldsymbol{W}_2$, and 1×3 for each hidden row.

(b) Remaining-loss functions and gradients. With the parameters and target fixed, the remaining-loss functions are

$$\begin{aligned} J_2(\boldsymbol{v}) &= (v_1 - y)^2, && \boldsymbol{v} \in \mathbb{R}^{1\times 1}, \\ J_1(\boldsymbol{v}) &= \left(\sum_{j=1}^{3} \mathrm{ReLU}(v_j) \cdot (\boldsymbol{W}_2)_{j1} + b_2 - y \right)^2, && \boldsymbol{v} \in \mathbb{R}^{1\times 3}. \end{aligned}$$

The derivative of J_2 with respect to its input coordinate, evaluated at the final preactivation row, is

$$[\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x}))]_1 = \left(\frac{\partial J_2}{\partial v_1} \right) (\boldsymbol{a}_2(\boldsymbol{x})) = 2 \cdot (f_{\boldsymbol{\theta}}(\boldsymbol{x}) - y) = -6.$$

Thus the initial backward row is $\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) = \begin{pmatrix} -6 \end{pmatrix}$. We now use this row and the saved hidden activations to obtain the output-layer parameter gradients:

$$\begin{aligned} \nabla_{\boldsymbol{W}_2} \mathcal{L}_{\mathrm{sq}}(\boldsymbol{\theta}) &= \boldsymbol{\phi}_1(\boldsymbol{x})^{\mathsf{T}} \cdot \nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) = \begin{pmatrix} -12 \\ -30 \\ 0 \end{pmatrix}, \\ \left(\frac{\partial \mathcal{L}_{\mathrm{sq}}}{\partial b_2} \right) (\boldsymbol{\theta}) &= -6. \end{aligned}$$

To apply (19), first calculate the row

$$\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) \cdot \boldsymbol{W}_2^{\mathsf{T}} = (-6) \cdot (2, -1, 1/2) = (-12, 6, -3).$$

The saved preactivations $\boldsymbol{a}_1(\boldsymbol{x}) = (2, 5, -2)$ give $\mathbf{step}(\boldsymbol{a}_1(\boldsymbol{x})) = (1, 1, 0)$. The componentwise product therefore gives

$$\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = (-12, 6, -3) \odot (1, 1, 0) = (-12, 6, 0).$$

The first layer receives the input $\boldsymbol{x} = (2, 0)$, so its parameter gradients are

$$\begin{aligned} \nabla_{\boldsymbol{W}_1} \mathcal{L}_{\mathrm{sq}}(\boldsymbol{\theta}) &= \boldsymbol{x}^{\mathsf{T}} \cdot \nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} -24 & 12 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \nabla_{\boldsymbol{b}_1} \mathcal{L}_{\mathrm{sq}}(\boldsymbol{\theta}) &= \nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = (-12, 6, 0). \end{aligned}$$

(c) Why some derivatives are zero. The negative third preactivation gives the multiplier $\text{step}(-2) = 0$. We therefore multiply the third coordinate -3 by zero when calculating $\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x}))$. The third hidden unit's first-layer weights and bias consequently have zero loss derivatives for this observation. Its output weight also has zero gradient because its activation is zero. The second row of the first weight gradient is zero for a different reason: each entry is multiplied by $x_2 = 0$.

(d) One gradient descent update. We now subtract η times each completed gradient from the parameter value used in the forward pass. All four updates use the gradients evaluated at the same $\boldsymbol{\theta}$:

$$\boldsymbol{W}_1^{\text{new}} = \begin{pmatrix} 1 + 24 \cdot \eta & 2 - 12 \cdot \eta & -1 \\ -1 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{b}_1^{\text{new}} = (12 \cdot \eta, 1 - 6 \cdot \eta, 0),$$

$$\boldsymbol{W}_2^{\text{new}} = \begin{pmatrix} 2 + 12 \cdot \eta \\ -1 + 30 \cdot \eta \\ 1/2 \end{pmatrix}, \qquad b_2^{\text{new}} = -1 + 6 \cdot \eta.$$

Updating $\boldsymbol{W}_2$ before calculating the first-layer gradients would mix two different parameter settings in one backward pass.

2. Binary classification.

(a) Forward pass and logistic loss. The first layer gives

$$\boldsymbol{a}_1(\boldsymbol{x}) = (1) \cdot \begin{pmatrix} 1 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 \end{pmatrix}, \qquad \boldsymbol{\phi}_1(\boldsymbol{x}) = \begin{pmatrix} 1 & 2 \end{pmatrix}.$$

The output logit is

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = 1 \cdot 2 + 2 \cdot (-1) + 0 = 0, \qquad \boldsymbol{a}_2(\boldsymbol{x}) = \begin{pmatrix} 0 \end{pmatrix}.$$

Consequently,

$$p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(0) = \frac{1}{2}, \qquad \mathcal{L}_{\log}(\boldsymbol{\theta}) = -\log \frac{1}{2} = \log 2.$$

(b) The output-layer derivatives. With the target $y = 1$ fixed, the remaining function is

$$J_2(\boldsymbol{u}) = \ell_{\log}(\sigma(u_1), 1) = -\log \sigma(u_1), \qquad \boldsymbol{u} \in \mathbb{R}^{1 \times 1}.$$

The initial backward row is

$$\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) = \begin{pmatrix} p_{\boldsymbol{\theta}}(\boldsymbol{x}) - y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} - 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \end{pmatrix}.$$

This already includes differentiation through the sigmoid. Using the saved hidden activations gives

$$\nabla_{\boldsymbol{W}_2} \mathcal{L}_{\log}(\boldsymbol{\theta}) = \boldsymbol{\phi}_1(\boldsymbol{x})^{\mathsf{T}} \cdot \nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -1 \end{pmatrix},$$

$$\frac{\partial \mathcal{L}_{\log}}{\partial b_2}(\boldsymbol{\theta}) = -\frac{1}{2}.$$

(c) The hidden-layer derivatives. Both hidden preactivations are positive, so $\mathbf{step}(\boldsymbol{a}_1(\boldsymbol{x})) = (1, 1)$. The preceding gradient is

$$\begin{aligned}\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) &= \left(\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) \cdot \boldsymbol{W}_2^{\mathsf{T}}\right) \odot \mathbf{step}(\boldsymbol{a}_1(\boldsymbol{x})) \\ &= \left(\left(-\tfrac{1}{2}\right) \cdot \begin{pmatrix} 2 & -1 \end{pmatrix}\right) \odot \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & \tfrac{1}{2} \end{pmatrix}.\end{aligned}$$

Since the input row is $\boldsymbol{x} = (1)$, the first-layer parameter gradients are

$$\nabla_{\boldsymbol{W}_1} \mathcal{L}_{\log}(\boldsymbol{\theta}) = \boldsymbol{x}^{\mathsf{T}} \cdot \nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & \tfrac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & \tfrac{1}{2} \end{pmatrix},$$

$$\nabla_{\boldsymbol{b}_1} \mathcal{L}_{\log}(\boldsymbol{\theta}) = \nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} -1 & \tfrac{1}{2} \end{pmatrix}.$$

3. Classification with three classes.

(a) Forward pass and cross-entropy loss. The first layer is unchanged, so $\boldsymbol{a}_1(\boldsymbol{x}) = \boldsymbol{\phi}_1(\boldsymbol{x}) = (1, 2)$. The output row is

$$\boldsymbol{a}_2(\boldsymbol{x}) = \begin{pmatrix} 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} + \begin{pmatrix} -1 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}.$$

All three exponentials in softmax equal one. Hence

$$\boldsymbol{p}_{\boldsymbol{\theta}}(\boldsymbol{x}) = \begin{pmatrix} \tfrac{1}{3} & \tfrac{1}{3} & \tfrac{1}{3} \end{pmatrix}, \qquad \mathcal{L}_{\text{multi}}(\boldsymbol{\theta}) = -\log \frac{1}{3} = \log 3.$$

(b) The output-layer derivatives. With the observed class $y = 2$ fixed, the remaining function is

$$\begin{aligned} J_2(\boldsymbol{u}) &= \ell_{\text{multi}}(\text{softmax}(\boldsymbol{u}), 2) \\ &= -\log\left(\frac{e^{u_2}}{e^{u_1} + e^{u_2} + e^{u_3}}\right), \qquad \boldsymbol{u} \in \mathbb{R}^{1\times 3}. \end{aligned}$$

Subtracting one from the probability of the observed class gives

$$\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) = \begin{pmatrix} \tfrac{1}{3} & \tfrac{1}{3} - 1 & \tfrac{1}{3} \end{pmatrix} = \begin{pmatrix} \tfrac{1}{3} & -\tfrac{2}{3} & \tfrac{1}{3} \end{pmatrix}.$$

Differentiation through softmax is already included in this row. The output-layer parameter gradients are

$$\begin{aligned}\nabla_{\boldsymbol{W}_2} \mathcal{L}_{\text{multi}}(\boldsymbol{\theta}) &= \boldsymbol{\phi}_1(\boldsymbol{x})^{\mathsf{T}} \cdot \nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) \\ &= \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} \tfrac{1}{3} & -\tfrac{2}{3} & \tfrac{1}{3} \end{pmatrix} = \begin{pmatrix} \tfrac{1}{3} & -\tfrac{2}{3} & \tfrac{1}{3} \\ \tfrac{2}{3} & -\tfrac{4}{3} & \tfrac{2}{3} \end{pmatrix},\end{aligned}$$

$$\nabla_{\boldsymbol{b}_2} \mathcal{L}_{\text{multi}}(\boldsymbol{\theta}) = \nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) = \begin{pmatrix} \tfrac{1}{3} & -\tfrac{2}{3} & \tfrac{1}{3} \end{pmatrix}.$$

(c) Adding the contributions from the output coordinates. The ordinary matrix product combines contributions from the three output coordinates:

$$\begin{aligned}\nabla J_2(\boldsymbol{a}_2(\boldsymbol{x})) \cdot \boldsymbol{W}_2^{\mathsf{T}} &= \begin{pmatrix} \tfrac{1}{3} & -\tfrac{2}{3} & \tfrac{1}{3} \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 \end{pmatrix}.\end{aligned}$$

In particular, its first entry is

$$\frac{1}{3} \cdot 1 + \left(-\frac{2}{3}\right) \cdot (-1) + \frac{1}{3} \cdot 0 = 1.$$

The second entry is $\frac{1}{3} \cdot 0 + (-\frac{2}{3}) \cdot 1 + \frac{1}{3} \cdot (-1) = -1$. Both hidden preactivations are positive. Applying the componentwise factors therefore gives

$$\nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} 1 & -1 \end{pmatrix} \odot \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

The negative second entry is retained because the second preactivation is positive. Finally,

$$\nabla_{\boldsymbol{W}_1} \mathcal{L}_{\text{multi}}(\boldsymbol{\theta}) = \boldsymbol{x}^\mathsf{T} \cdot \nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix},$$

$$\nabla_{\boldsymbol{b}_1} \mathcal{L}_{\text{multi}}(\boldsymbol{\theta}) = \nabla J_1(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} 1 & -1 \end{pmatrix}.$$